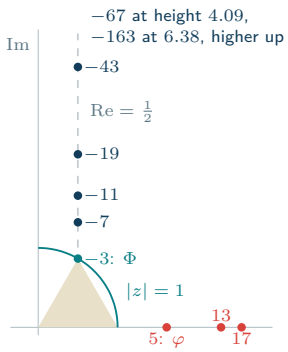
NUMBER THEORY ON ONE PAGE

Golden-type numbers, real and complex

The recipe $(1+\sqrt{d})/2$ gives φ at $d=5$ and Φ at $d=-3$, and reaches Euler's prime polynomial at $d=-163$.

1 / THE RULE AND THE PICTURE

The golden ratio and its complex companion are the first members of one family



For a squarefree whole number $d \equiv 1 \pmod 4$ put $\tau_d = \frac{1+\sqrt{d}}{2}$. It solves a quadratic with whole coefficients,

$$\tau^2 - \tau + k = 0, \qquad k = \tfrac{1-d}{4} = \tau\bar{\tau}$$

and the numbers $a + b\tau_d$ with whole a, b form the ring of integers of $\mathbb{Q}(\sqrt{d})$ (the ring $\mathbb{Z}[\tau_d]$). The real ones, $d > 0$, lie on the real axis, with φ at $d = 5$.

The complex ones, $d < 0$, lie on the vertical line $\mathrm{Re} = \frac{1}{2}$ at height $\frac{\sqrt{|d|}}{2}$, with $\Phi = e^{i\pi/3}$ at $d = -3$, the one point that is also on the unit circle, where the triangle $0, 1, \Phi$ is equilateral.
Every other τ_d has $\tau\bar{\tau} = |k| > 1$.

2 / THE FAMILY

The constant term k grows as the point climbs the line $\mathrm{Re} = \frac{1}{2}$

d	τ_d	τ_d solves	$k = N(\tau_d)$	units of $\mathbb{Z}[\tau_d]$	$a^2 - a + k$ prime for
5	$\varphi = (1+\sqrt{5})/2$	$x^2 - x - 1$	-1	$\pm\varphi^n$, infinitely many	–
13	$(1+\sqrt{13})/2$	$x^2 - x - 3$	-3	infinitely many	–
-3	$\Phi = (1+\sqrt{-3})/2$	$x^2 - x + 1$	1	six: the powers of Φ	–
-7	$(1+\sqrt{-7})/2$	$x^2 - x + 2$	2	± 1	$a = 0, 1$
-11	$(1+\sqrt{-11})/2$	$x^2 - x + 3$	3	± 1	$a = 0, \ldots, 2$
-19	$(1+\sqrt{-19})/2$	$x^2 - x + 5$	5	± 1	$a = 0, \ldots, 4$
-43	$(1+\sqrt{-43})/2$	$x^2 - x + 11$	11	± 1	$a = 0, \ldots, 10$
-67	$(1+\sqrt{-67})/2$	$x^2 - x + 17$	17	± 1	$a = 0, \ldots, 16$
-163	$(1+\sqrt{-163})/2$	$x^2 - x + 41$	41	± 1	$a = 0, \ldots, 40$

The last column lists the values a for which $a^2 - a + k$ is prime. For every one of these k the values are prime for exactly $0 \le a < k$ and first fail at $a = k$, since $k^2 - k + k = k^2$.

3 / THE TWINS: $x^2 - x - 1$ AND $x^2 - x + 1$

Only φ and Φ are units, and they differ by one sign

The number τ_d is a unit exactly when $k = \pm 1$, which for $d \equiv 1 \pmod 4$ happens only for $d = 5$ and $d = -3$, the polynomials $x^2 - x - 1$ and $x^2 - x + 1$.
The real unit φ has infinite order, so φ^n grows and the units $\pm\varphi^n$ are infinitely many, while the complex unit Φ has $|\Phi| = 1$ and order 6: its powers are the six units of the Eisenstein integers, and $\Phi = 1 + \omega = -\omega^2$ with $\omega = e^{2\pi i/3}$.
For every other $d < 0$ the only units are ± 1. Growth against rotation is the hyperbola against the circle of the previous sheet.

4 / EULER'S POLYNOMIAL

The norm of $a - b\tau$ is $a^2 - ab + kb^2$, and at $b = 1$ it is $a^2 - a + k$

Because $\tau + \bar{\tau} = 1$ and $\tau\bar{\tau} = k$, the product $(a - b\tau)(a - b\bar{\tau}) = a^2 - ab + kb^2$, so $a^2 - a + k$ is the norm of $a - \tau$. For $k = 41$ it is prime for $a = 0, 1, \ldots, 40$.
The reason is that for $a \le 40$ the value is below $41^2 = 1681$, so a composite value would have a prime factor below 41, yet $p \mid a^2 - a + 41$ forces $(2a-1)^2 \equiv -163 \pmod p$, and -163 is a non-residue modulo every odd prime below 41 (and the value is always odd).
That every $p < k$ is inert is the unique-factorisation property of $\mathbb{Z}[\tau_d]$, and by the Baker–Heegner–Stark theorem $d = -3, -7, -11, -19, -43, -67, -163$ are the only such golden-type values.

ASIDE / A NEAR-INTEGER

$e^{\pi\sqrt{163}} = 262\,537\,412\,640\,768\,743.999\,999\,999\,999\,25\ldots$, within 7.5×10^{-13} of a whole number.

LINKS TO OTHER SHEETS

Golden and metallic sides; adding a new number; the Eisenstein and cyclotomic sheets.

Convention. $\tau_d = \frac{1+\sqrt{d}}{2}$; $k = \frac{1-d}{4} = N(\tau_d)$; $\omega = e^{2\pi i/3}$; prime-value claims were checked for every listed k.

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